

HVAC SYSTEM

The main components of a centralized HVAC (Heating, Ventilation, and Air Conditioning) cooling system:

Zone → AHU → Chiller → Cooling Tower.

Here's a clear, structured explanation of how these pieces fit together:

How a Zone–AHU–Chiller–Cooling Tower HVAC System Works

A typical large-building cooling system uses a central chilled-water plant with multiple zones. The cooling chain works from the building spaces out to the mechanical plant.

1. Zones (Conditioned Spaces)

These are the areas in the building that need temperature and humidity control (offices, rooms, halls, etc.).

Each zone usually has:

The zone's job:

➡ Request cooling as needed.

2. AHU (Air Handling Unit)

- ★ The AHU supplies conditioned air to zones.

What the AHU does: VAV box (Variable Air Volume terminal)

Fan coil unit (FCU) OR is fed directly by the AHU

Temperature sensors / thermostats

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- ★ Draws in return air + fresh outdoor air

- ★ Filters it

- ★ Passes it over a cooling coil (supplied with chilled water from the chiller)

- ★ Delivers cool air to zones via ducts

Key AHU components:

- ★ Supply fan
- ★ Return fan
- ★ Cooling coil (uses CHILLED WATER)
- ★ Heating coil (optional)
- ★ Filters
- ★ Dampers (OA/RA/EA)
- ★ AHU does not produce cooling — it uses chilled water that comes from the chiller.

3. Chiller (Chilled Water Plant)

The chiller produces chilled water (typically 6–8°C or 42–46°F) and sends it to AHUs.

Main chiller cycle:

Evaporator: chills the water sent to AHUs

Compressor: raises refrigerant pressure

Condenser: rejects heat to condenser water

Expansion valve: reduces pressure to restart refrigerant cycle

The chiller removes heat from the building (via chilled water) and dumps that heat into the condenser water loop, which then goes to the cooling tower.

4. Cooling Tower

The cooling tower rejects heat from the chiller to the atmosphere.

What happens:

- ★ Hot condenser water from the chiller enters the cooling tower
- ★ Air + water mix → evaporation removes heat

- ★ Cooler water returns to the chiller condenser
- ★ Typical condenser water temperatures:
- ★ Hot out to tower: $\sim 32\text{--}35^{\circ}\text{C}$ ($90\text{--}95^{\circ}\text{F}$)
- ★ Cooled return from tower: $\sim 27\text{--}30^{\circ}\text{C}$ ($80\text{--}86^{\circ}\text{F}$)

Overall System Flow Summary

A. Cooling Delivery Path

Building Zones

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AHU (uses chilled water to cool air)

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Chilled Water Loop

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Chiller (absorbs heat from chilled water)

B. Heat Rejection Path

Chiller Condenser

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Condenser Water Loop

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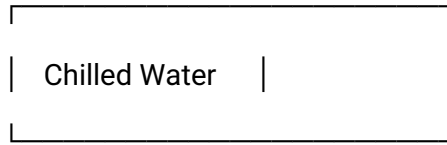
Cooling Tower (rejects heat to atmosphere)

[Zone Thermostat]

↓

[AHU Unit]

↓ (Air)



↓

[Chiller]

↓ (Condenser Water)

[Cooling Tower]

↓ (Heat to Atmosphere)